

INSURANCE FRAUD INVESTIGATION AGENT

Project Report

Submitted by

Mohammad Farzan Nawaz Faruqui

IIT Bhilai — Data Science & Analytics

GitHub Repository

<https://github.com/Farzan04/Insurance-Fraud-Investigation-Agent>

1. Executive Summary

This project presents an AI-powered insurance fraud investigation prototype that combines multi-source machine learning with a LangGraph-based investigation agent. The system assists an investigator by bringing together claim, policy, payment, and agent/ghost-broking information rather than relying only on claim-level attributes.

A Multi-Source Random Forest is used as the quantitative fraud-prediction component. The investigation agent retrieves verified information through deterministic tools, obtains the ML fraud probability, and produces an evidence-based assessment containing key evidence, mitigating or contradictory evidence, a recommendation, and a final classification.

The final model was evaluated on a 200-claim held-out split. It achieved ROC-AUC 0.7096, fraud precision 0.4688, fraud recall 0.4054, and fraud F1 0.4348. A separate 10-claim development-sample evaluation of the ML-backed agent produced 90% accuracy, 100% fraud precision, 80% fraud recall, and 88.89% fraud F1; this small sample is treated as development evaluation rather than a held-out test result.

2. Objectives

- Build a fraud-prediction model using multiple related insurance data sources.
- Compare cross-source information with a claim-only baseline.
- Create deterministic tools for claim, policy, payment, and agent-related evidence retrieval.
- Integrate the tools into a LangGraph investigation workflow.
- Separate ML prediction from the broader evidence-based investigation assessment.
- Provide an end-to-end prototype for investigating a supplied claim ID.

3. Problem Statement

Insurance fraud investigation can require information distributed across several records. A claim may look ordinary in isolation while related policy, payment, customer, or agent information may provide additional context.

The project therefore treats fraud investigation as a combination of prediction and evidence retrieval. The prototype is decision support for an investigator, not an autonomous claim-adjudication system.

4. Data and Ground Truth

The project uses synthetic datasets. Final validation reports 1,000 records each for claims, policies, payments, and agent/ghost-broking records. A multi-source dataset is constructed by joining related records.

Data source	Size	Purpose
Claims	1,000 × 17	Claim attributes and fraud target
Policies	1,000 × 16	Policy-level context
Payments	1,000 × 14	Payment and premium information
Agent / Ghost-broking	1,000 × 14	Agent-related risk context

4.1 Ground-Truth Definition

The quantitative ML evaluation uses the dataset's `fraud_flag` field as the ground-truth target. During agent investigation, `fraud_flag` and `fraud_type` are deliberately hidden from the investigation tools so that the agent cannot simply read the answer. They are retained separately for model training and evaluation.

This separation is important: the model can be evaluated against known labels, while the agent's investigation remains evidence-driven.

5. Methodology

5.1 Data Preparation and Feature Engineering

- Claims are used as the starting point for the multi-source feature table.
- Policy, payment, and agent/ghost-broking information are joined using appropriate identifiers.
- `fraud_flag` is the target; `fraud_flag` and `fraud_type` are removed from the model feature matrix.
- Numeric and categorical variables are preprocessed with imputation and one-hot encoding.
- The final multi-source model uses 29 features.

5.2 Machine Learning

The final quantitative model is a Multi-Source Random Forest. The final validation reports 800 training rows and 200 held-out test rows. The model outputs a fraud probability and predicted class through `fraud_prediction_tool`.

5.3 Investigation Tools

- `claim_tool` — retrieves claim information.
- `policy_tool` — retrieves policy information.
- `payment_tool` — retrieves payment records.
- `agent_tool` — retrieves agent/ghost-broking records.
- `evidence_tool` — prepares deterministic investigation evidence.
- `investigation_context_tool` — combines verified context for a claim.
- `fraud_prediction_tool` — applies the trained Random Forest and returns probability/class.

5.4 LangGraph Investigation Agent

The agent is implemented with LangGraph and uses Qwen as the LLM component. The workflow retrieves relevant context, calls the ML prediction tool, and generates an evidence-based assessment. The agent instructions explicitly prevent access to the hidden ground-truth labels.

6. System Architecture

The end-to-end flow is:

Claim ID supplied by investigator

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Investigation context: Claim → Policy → Payment → Agent/Ghost-broking → Evidence

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Fraud prediction: Multi-Source Random Forest → probability + class

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LangGraph investigation agent

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Key evidence + mitigating evidence + recommendation + final classification

7. Model Evaluation

The fraud model is evaluated on a held-out test split using ROC-AUC, fraud precision, fraud recall, and fraud F1.

Model / evaluation	Metric	Result
Claim-only baseline	ROC-AUC	0.6783
Multi-source Random Forest	ROC-AUC	0.7096
Multi-source Random Forest	Fraud Precision	0.4688
Multi-source Random Forest	Fraud Recall	0.4054
Multi-source Random Forest	Fraud F1	0.4348
Relationship-enhanced model	ROC-AUC	0.6801

The multi-source ROC-AUC increased from 0.6783 for the claim-only baseline to 0.7096, indicating useful fraud-related signal from additional cross-source information.

8. Ground Truth, False Positives, and Validation

A predicted fraud case can only be assessed as a correct or incorrect prediction by comparing it with the known ground-truth label. The project does this for the ML evaluation using fraud_flag on the held-out test set.

- Fraud precision = 0.4688: among claims predicted as fraud, about 46.88% were fraud according to the ground-truth labels.
- Fraud recall = 0.4054: about 40.54% of the ground-truth fraud cases were detected.
- Fraud F1 = 0.4348: harmonic mean of fraud precision and recall.
- ROC-AUC = 0.7096: measures discrimination across probability thresholds.

Thus the project does not claim that every positive prediction is correct. The precision and recall values quantify the model's false-positive and false-negative behavior on the held-out test set.

9. Agent Evaluation

The LangGraph-based investigation agent was tested on a 10-claim development sample. Both the investigation-context tool and fraud-prediction tool were used successfully in 100% of those cases.

Development metric	Result
Accuracy	90.00%
Fraud Precision	100.00%

Fraud Recall	80.00%
Fraud F1	88.89%
Context tool usage	100%
Prediction tool usage	100%

These results are development-sample results only because the sample contains 10 claims and is not a held-out test set.

10. End-to-End Demonstration

The final end-to-end test uses claim CLM000001. The investigation context is retrieved, the ML prediction is obtained, and the agent produces an evidence-based assessment.

Test item	Result
Test claim	CLM000001
Investigation context	SUCCESS
Fraud probability	0.21
ML classification	NON-FRAUD
Final agent assessment	SUCCESS

For this demonstration claim, the model predicted a 21% fraud probability and classified the claim as NON-FRAUD. The agent also reported contradictory or mitigating evidence and recommended further review, illustrating the distinction between model prediction and investigation assessment.

11. Key Findings

- Multiple related data sources improved ROC-AUC over the claim-only baseline in the reported experiment.
- The Multi-Source Random Forest provides a quantitative fraud probability consumed by the agent.
- Investigation tools retrieve verified information without exposing the ground-truth fraud labels.
- LangGraph connects evidence retrieval, ML prediction, and LLM-based assessment.
- The system is a decision-support prototype rather than an autonomous adjudication system.

12. Limitations

- The datasets are synthetic.
- The ML evaluation uses a 200-claim held-out split.
- The agent development evaluation uses only 10 claims and is not a held-out test.
- The manually designed evidence score was not sufficiently discriminative to serve as the final classifier.
- LLM classification was not used as the primary quantitative performance measure because it could abstain with UNCERTAIN.
- Results should therefore be interpreted as prototype evaluation rather than production-level fraud detection performance.

13. Future Scope

- Evaluate on larger, real-world and properly governed insurance datasets.
- Tune classification thresholds according to the business cost of false positives and false negatives.
- Perform stronger probability calibration and cross-validation.

- Add SHAP or similar explainability for investigator-facing model explanations.
- Expand relationship/graph analysis among customers, policies, agents, and claims.
- Add audit logging and human approval before operational decisions.
- Benchmark additional models and imbalance-aware training approaches.

14. Repository and Project Files

The complete project repository is available at:

<https://github.com/Farzan04/Insurance-Fraud-Investigation-Agent>

The repository contains:

- Insurance_Fraud_Agent.ipynb — complete Jupyter Notebook.
- data/ — claims, ghost-broking, payment, and policy CSV datasets.
- presentation/ — project presentation.
- results/ — model comparison and fraud F1 visualizations.
- README.md — project overview, architecture, run instructions, outputs, and limitations.

15. Conclusion

The Insurance Fraud Investigation Agent demonstrates how traditional machine learning and an agentic workflow can be combined for insurance fraud investigation support. The Random Forest supplies a measurable fraud prediction component, while LangGraph coordinates evidence retrieval and reasoning over multiple sources.

The resulting prototype can investigate a claim end-to-end while keeping ground-truth labels separate from the investigation process. Its reported results should be interpreted in the context of synthetic data and limited evaluation size, with further validation required before production deployment.